

CryptoTrace I4C

Automated Crypto Fund-Flow Tracing & VASP Identification



Problem Statement ID – SIH26183



Problem Statement Title – Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics



Theme – Blockchain & Cybersecurity



PS Category – Software



Team ID – To be updated after selection



Team Name – Innovative Tech



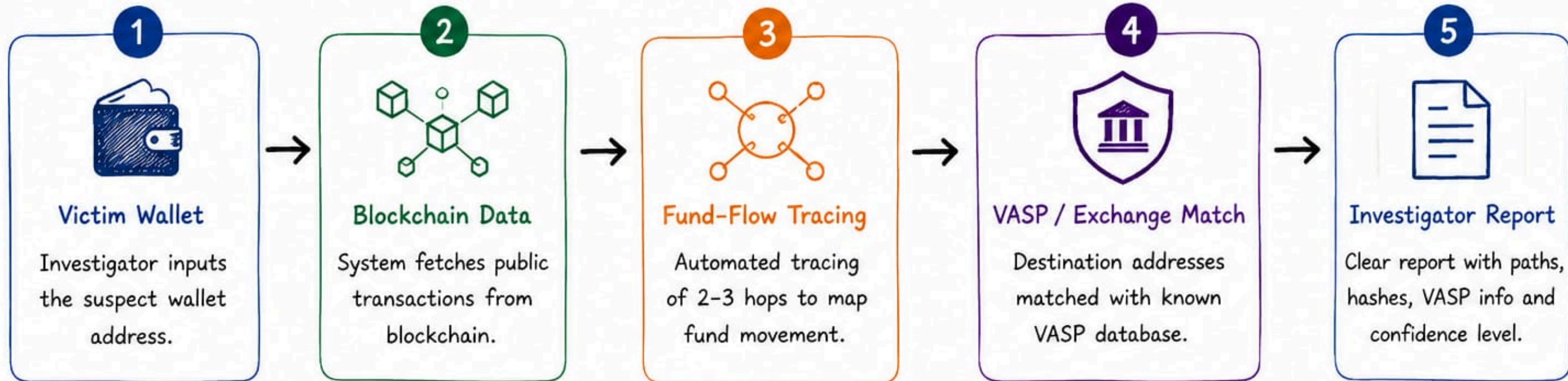


CryptoTrace I4C

Our proposed solution in one simple flow



HOW IT WORKS



WHAT THIS SOLVES



- Cuts down manual searching across multiple explorers.
- Shows the complete fund path and key intermediary wallets.
- Identifies the exchange/VASP where funds are likely to land.

KEY BENEFITS



- Faster investigations
- Evidence ready for legal action
- Easy to understand and explain
- Built using real public blockchain data

UNIQUE EDGE



- Deterministic tracing (no black-box AI)
- VASP matching with confidence score
- Simple, transparent and investigator-first



TECHNICAL APPROACH

How we turn a wallet address into an investigation trail



1) TECHNOLOGY STACK



Frontend

- React
- Cytoscape.js

Interactive graph visualization for transactions.



Backend

- Python
- FastAPI
- NetworkX

API service and graph processing engine.



Blockchain Data

- Etherscan API (Ethereum)
- TronScan API (Tron)

Fetch real-time public transactions.



Data & Matching

- VASP address dataset
- Entity matching
- Risk / confidence scoring

Match destinations with known VASP and score confidence.



Security

- Secure API usage
- Data validation

Protect data, handle errors and ensure reliability.

2) IMPLEMENTATION FLOW



1. Wallet Address
Victim provides the suspect wallet address.



2. Fetch Transactions
Collects transactions from blockchain APIs.



3. Build 2-3 Hop Graph
Creates a simple transaction graph (2-3 hops).



4. Fund-Flow Tracing
Analyzes the graph to trace fund movement.



5. VASP Match
Match destination addresses with VASP dataset.



6. Evidence / Report
Generate report with paths, VASP info and confidence score.

3) 3-5 DAY MVP (WHAT WE BUILT FIRST)



ETH + TRON

Support for Ethereum and Tron networks.



Interactive Graph

Visual view of transactions and fund flow.



VASP Confidence

Confidence score for matched VASP.



PDF Report

Investigator-ready report with evidence and details.

✓ ANALYSIS OF FEASIBILITY

The solution is practical and achievable with available tools, public data and a focused scope.



Public blockchain data & APIs (Etherscan, TronScan) are openly available.



Bounded 2–3 hop tracing keeps the system light, fast and easy to demonstrate.



Small, source-labelled VASP dataset ensures transparent and auditable attribution.



Human / fiat fusion is demonstrated using labelled synthetic data for demo purpose.



POTENTIAL CHALLENGES & RISKS

Some factors can affect accuracy, coverage or real-world usage.



API limits / outages

Rate limits or downtime in public APIs may affect data retrieval.



Stale or unknown VASP labels

New exchange addresses or unlabelled wallets can reduce attribution accuracy.



Heuristic false positives

Pattern-based rules can sometimes flag normal transactions as risky.



Mixer / bridge boundaries

Privacy tools and cross-chain bridges can break the trace.



Private KYC not visible on-chain

We cannot see real user identity or KYC data on the blockchain.



Real bank / telecom data access

Live data from banks or telecom requires proper legal authorisation.



STRATEGIES FOR OVERCOMING

We handle risks using the right approach and clear boundaries.



Cache data + backup

Maintain local cache and backup APIs for critical calls.



Show rule + confidence

Every match shows confidence level. Heuristics are leads, not proof.



Store labelled VASP data

Every VASP label includes source name and last updated date.



Flag unknowns

Clear labels: "Unknown / Not in DB" to avoid over-claiming.



Human review & phased scope

Human review before legal action. Expand scope after MVP stability.



OUR APPROACH

We keep the system transparent, explainable and investigator-first.

The goal is to support better decisions — not to replace investigators or legal procedures.



Collect data



Trace & analyse



Score & match



Generate report



Human review

POTENTIAL IMPACT

CryptoTrace I4C brings wallet activity, transaction paths and VASP information into one place. Instead of manually checking multiple explorers, investigators can follow the money from the reported wallet and reach important endpoints faster.

It helps investigators to:

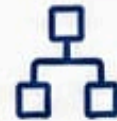
- Trace the movement of suspected fraud funds across 2-3 hops.
- Highlight important intermediary wallets in the fund flow.
- Match destination addresses with known VASP/exchange addresses.
- Generate a structured evidence trail for further investigation.

KEY BENEFITS



1. Faster Investigation

Reduces repetitive manual searching across explorers and helps investigators reach relevant transaction paths quickly.



2. Clear Fund-Flow View

Shows how funds move between wallets instead of presenting isolated transaction records.



3. VASP Identification

Highlights where traced funds reach an address associated with a known VASP/exchange, along with a confidence level.



4. Better Evidence Preparation

Provides transaction hashes, wallet addresses, fund-flow paths and matching information in one report.



5. Easy to Extend

The same approach can be extended to additional blockchains, address labels and authorised data sources.

REAL-WORLD VALUE



Victim Report

Fraud is reported with wallet details



Faster Tracing

Automated tracing of fund movement



Clearer Evidence

Structured and verifiable evidence



Better Lead

Stronger direction for investigation

This helps agencies take informed action quickly and focus on real leads.



BOUNDARY / IMPORTANT LIMITATION

- CryptoTrace I4C identifies investigative leads, not people.
- A wallet / VASP match does not automatically prove the identity or guilt of a person. Final identification and legal action remain with authorised investigators and the appropriate legal process.
- CryptoTrace I4C supports the investigation process; it does not replace human judgement or legal procedures.
- Output is based on available public data, labels and rules, which may change over time.



DETAILS / LINKS OF REFERENCE AND RESEARCH WORK

- SIH 2026 Problem Statement SIH26183 — Ministry of Home Affairs / I4C.
- We built CryptoTrace I4C to automate fund-flow tracing, identify linked exchanges (VASPs) and prepare investigator-ready reports.
- Our approach combines public blockchain data, graph analytics and labelled VASP data to give investigators a single, connected view.



REFERENCE FILES / DOCUMENTS

- SIH Idea Submission Template (v1.0)
- Public VASP addresses dataset (source-labelled)
- Etherscan API Documentation
- TronScan API Documentation
- NetworkX Documentation
- Cytoscape.js Documentation
- FastAPI Documentation



KEY RESEARCH POINTS USED

- Existing tools (e.g., Etherscan, Tronscan, Chainalysis Reactor) provide tracing and analytics, but require manual steps across multiple explorers.
- Graph analytics helps reveal fund movement and cluster relationships.
- VASP / exchange labels and heuristics improve attribution confidence.
- MVP scope: Ethereum + TRON, 2–3 hops, source-labelled VASP data, graph view, confidence scoring and report generation.



ACADEMIC & TECH REFERENCES (EXAMPLES)

- Meiklejohn, S., Pomarole, M., Jordan, G., Levchenko, K. (2013). A fistful of bitcoins: Characterizing payments among men with no names. Proceedings of the Internet Measurement Conference (IMC).
- Chainalysis. (2022). 2022 Crypto Crime Report.
- Elliptic. (2023). Cryptocurrency Anti-Money Laundering Report 2023.
- Zhang, Y., et al. (2019). Tracking suspicious transactions in blockchain using graph analysis. IEEE Access.
- Buterin, V. (2014). A next-generation smart contract and decentralized application platform. Ethereum Whitepaper.



OUR APPROACH TO RESEARCH



1. Study & Understand

Read SIH brief, research papers and existing tools to understand the gaps.



2. Collect & Test Data

Collect public transactions and test APIs to validate the core idea.



3. Analyse & Build

Build graph models, apply heuristics and test on real cases.



4. Evaluate & Iterate

Check results with known cases, refine rules and improve accuracy.



5. Document & Share

Document our method and share references, data sources and limitations.